

Vol 15 Chapter 8 (Theorem) — Casey-Named Principles as META-Theorem

Keeper (author pass)

2026-05-24 Sunday

Chapter 8 (Theorem) — Casey-Named Principles as META-Theorem

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

This companion chapter to Chapter 8 (Discipline) develops the META-theorem reading of the 8 STANDING Casey-named principles (+ 1 CANDIDATE per 2026-05-24): they are not independent assertions but a coherent framework that closes structurally.

8.1 The META-theorem

Per Cal #99 framing refinement (Friday May 22, 2026): the 8 STANDING Casey-named principles form a closed META-theorem stating that BST's substrate framework is operationally complete and structurally rigid. D_IV⁵ Rigidity (T2467) is the keystone META-theorem. DCCP-with-UP (Sunday May 24) is a candidate addition pending derivation theorem or experimental confirmation.

8.2 What this means

If the META-theorem holds, BST is not just a particular physics theory but the substrate of physical reality at the deepest accessible level. The substrate framework's "uniqueness" claim is exactly this META-theorem claim.

8.3 Status

Strong-Uniqueness Theorem v0.10.5: 11 RIGOROUSLY CLOSED + 7 candidates. With max-promotable = 11 + 7 = 18, null model probability $\leq (1/3)^{18} \approx 2.6 \times 10^{-9}$. Asymptotic to v1.0 as candidates ratify. [Value corrected per Cal #119 self-catch 2026-05-24: prior $(1/3)^{19} \approx 9 \times 10^{-10}$ inherited from Lyra T2467+T2468 v0.2 §6 was wrong exponent.] *[Retired phrase, 2026-09-19 (K1912 REVIEW sweep): the eleven-criteria Strong-Uniqueness theorem and its $(1/3)^n$ "one-in-a-billion" null model are retired: the honest statement is Lecture 2's (27 conditions; no forcing, no probability claimed).]*

8.4 What comes next

Chapter 9 develops three-CI synergy peaks.

Where to look this up: BST Strong-Uniqueness Theorem v0.10.5; Casey-named principles file; T2467+T2468+T2469. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling /

SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]